

Task 1 - Selection and Research

Introduction

Banks are increasingly relying on AI-driven credit scoring systems to process loan applications more efficiently. These systems can analyse large volumes of applications quickly and reduce manual review costs. However, once a model is deployed across multiple jurisdictions, compliance becomes more complicated. The same model behaviour may be considered acceptable in one country but problematic in another. This project focuses on how regulatory expectations diverge between the United States and Singapore when monitoring AI-based retail credit scoring systems.

1. Selected Entity: Standard Chartered Bank

Standard Chartered Bank was selected because it operates across multiple jurisdictions, including Singapore, the United States, Hong Kong, and the United Kingdom. This makes it an appropriate case for studying cross-border AI governance problems. As a global bank, Standard Chartered must comply with different regulatory expectations regarding fairness, explainability, and model governance. The bank has also invested heavily in digital banking and AI-enabled financial services, making AI credit scoring a realistic use case for this project.

2. Selected Domain: AI Governance in Retail Credit Scoring

The selected domain for this project is AI governance in retail credit scoring. Retail credit scoring models are commonly used to determine whether applicants should be approved for loans or credit products. These models evaluate factors such as income, debt-to-income ratio, employment status, and credit history. A model trained during one economic environment may become less reliable when macroeconomic conditions change.

3. Dataset Background

The model SCB-CS-001 was trained using historical loan applications from 2022-2023 and later deployed in a live environment during 2025-2026. During this period, applicant income declined, loan amounts increased, unemployment rose, and the proportion of low-score applicants became larger. These changes suggest that the live deployment environment no longer resembles the original training environment, creating a clear model drift problem.

4. Regulatory Divergence: OCC vs MAS

Area	OCC / United States	MAS / Singapore
Main focus	Fair lending	AI governance
Key concern	Consumer harm	Model oversight
Priority	Approval fairness	Drift monitoring

Explainability	Strong emphasis	Moderate emphasis
Human oversight	Recommended	Strongly emphasised
Compliance trigger	Demographic disparity	Governance deterioration

The core idea behind this project is that the same AI model may trigger different compliance concerns depending on the jurisdiction. In the United States, regulators such as the OCC place stronger emphasis on fair lending, consumer protection, adverse action explanation, and discrimination risk. In Singapore, MAS places stronger emphasis on governance, oversight, operational accountability, and responsible AI deployment.

5. Why RegLens AI Is Necessary

A major challenge for multinational banks is that compliance requirements cannot be fully standardised across jurisdictions. The same dataset may produce a fairness concern in the United States but a governance escalation issue in Singapore. This project therefore proposes RegLens AI, a hypothetical jurisdiction-aware RegTech monitoring tool capable of monitoring live model behaviour, detecting distribution drift, evaluating fairness indicators, and generating jurisdiction-specific compliance alerts.

6. Project Objective

The objective of this project is to design RegLens AI as a jurisdiction-aware compliance monitoring prototype for the SCB-CS-001 retail credit scoring model. The tool compares historical training data against live deployment data to identify distribution drift, performance deterioration, demographic outcome disparities, and governance escalation triggers.

References

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